

example, let us consider the SGX543, it is immensely popular and is licensed by a lot of companies for their products but there are 5 variants ranging from single core to a model with sixteen cores. Have a look as to how much their processing power differs when benchmarked at the same clock speed.

- ▶ (single core) 35M polygon/s @200 MHz
- ▶ (two cores) 68M polygon/s @200 MHz
- ▶ (four cores) 133M polygon/s @200 MHz
- ▶ (eight cores) 266M polygon/s @200 MHz
- ▶ (sixteen cores) 532M polygon/s @200 MHz

The performance ratio between the low-end and high-end variant is a good 500 million polygons per second. So if you that a certain smartphone has the SGX543 it may not necessarily have the 16 core variant. We do not have such benchmarks for every variant of every GPU, so consider these benchmarks purely as a reference.

The lack of cross-platform benchmarking tools is a pain in the sense that you cannot properly compare two devices having different OS against each other. There are plenty of cross platform browser benchmark tools, then again these won't paint the true picture at all. We have decided to used proceed with a tool that is available on all three major platforms - GFXBench. Now all of these benchmarks are offscreen benchmarks. By offscreen, we mean that if these GPUs were pitted against each other on a level playing field. In this set of benchmarks, we normalise all display sizes to a 1080p display and have the GPU perform the benchmark for that resolution (1920 x 1080). If you wish to see how different smartphones/tablets compare against each other then you have to consider the onscreen benchmarks.

Since more and more old PC games are being ported to the smartphone/tablet platform (although not officially), we are pretty sure some of you might be interested to see how SoC GPUs perform against mainstream desktop GPUs, Anandtech has done a wonderful article on the very subject.

<http://dgit.in/14loF2K> 